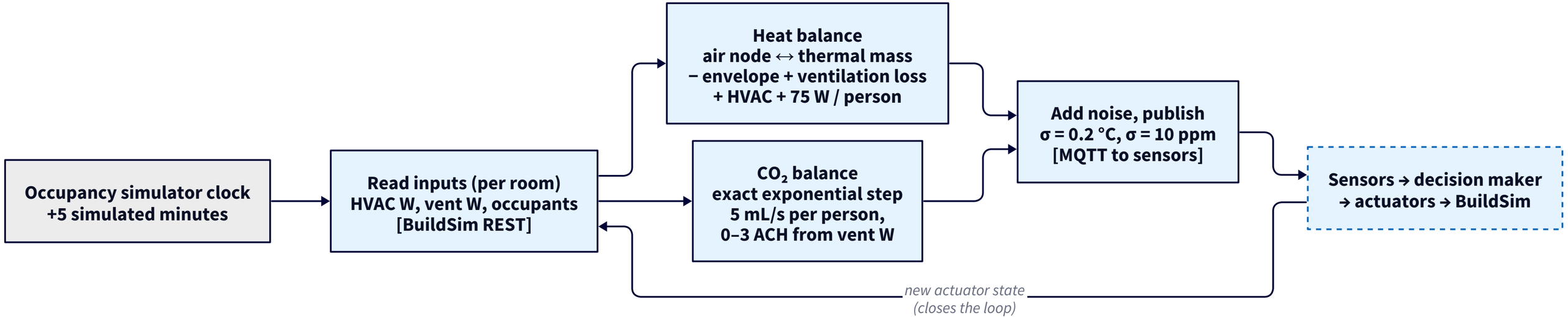

Occupancy simulator clock
+5 simulated minutes
Read inputs (per room)
HVAC W, vent W, occupants
[BuildSim REST]
Heat balance
air node ↔ thermal mass
− envelope + ventilation loss
+ HVAC + 75 W / person
CO₂ balance
exact exponential step
5 mL/s per person,
0–3 ACH from vent W
Add noise, publish
σ = 0.2 °C, σ = 10 ppm
[MQTT to sensors]
Sensors → decision maker
→ actuators → BuildSim
new actuator state
(closes the loop)